
0.1 Question 1a

Based on the columns in this dataset and the values that they take, what do you think each row represents? That is, what is the **granularity** of this dataset?

Each row represents a unique property in Cook County with detailed characteristics about the property itself, its location, features, sale information, assessments, and so on. That is, the granularity of this dataset is at the level of each of the property, where each record represents a property with a unique PIN, and each property is a single unit of real estate that is uniquely identified and tracked by its unique Permanent Identification Number.

0.2 Question 1b

Why was this data collected? For what purposes? By whom?

You should watch [Lecture 13](#) before attempting this question.

The data was collected in response to calls for a fairer tax system after scandals revealed problems with the old approach. Collected by the Cook County Assessor’s Office, it aims to produce accurate, market-based property assessments so that tax burdens are fairly allocated among property owners.

Because “equitable” is difficult to pin down—especially when deciding how to distribute taxes across groups with different income levels in a socially acceptable way—the project emphasizes handling that complexity and providing stronger justifications. It is part of a broader effort to correct systemic bias and past inaccuracies uncovered by investigations. Through the Open Data Initiative, the process is now more transparent, enabling the public and other stakeholders to examine and challenge property valuations.

0.3 Question 1c

Craft at least two questions about housing in Cook County that can be answered with this dataset and provide the type of analytical tool you would use to answer it (e.g. “I would create a ____ plot of ____ and ” **or** ”*I would calculate the* [summary statistic] for ____ and ____”). Be sure to reference the columns that you would use and any additional datasets you would need to answer that question.

1. O’Hare, land size, and roads: Among parcels within a defined radius of O’Hare, how do Land Square Feet and proximity to major roads relate to Sale Price, after accounting for structure and time? I would fit a hedonic multiple linear regression on $\log(\text{Sale Price})$ using Land Square Feet, a derived Road Proximity measure (distance to nearest primary/arterial), and controls like Building Square Feet, Age, Repair Condition, Central Air, Neighborhood Code, and Sale Quarter; columns used: Sale Price, Land Square Feet, Building Square Feet, Age, Repair Condition, Central Air, Neighborhood Code, Sale Quarter, plus derived Road Proximity; additional datasets: OpenStreetMap (road centerlines) and FAA noise contours to optionally add an O’Hare Noise index.
2. Does having central air conditioning correspond to higher sale prices in Cook County? Analytical Tool: I would first compute and compare the mean Sale Price for homes with vs. without Central Air and perform a t-test to assess statistical significance; then, for a more controlled estimate, I’d fit a hedonic regression of $\log(\text{Sale Price})$ on Central Air while controlling for structure, location, and time (e.g., Building Square Feet, Land Square Feet, Age, Repair Condition, Fireplaces, Neighborhood Code, Sale Quarter). Columns Used: Sale Price, Central Air (plus listed controls). Additional Datasets: None required.

0.4 Question 1d

Suppose now, in addition to the information already contained in the dataset, you also have access to several new columns containing demographic data about the owner, including race/ethnicity, gender, age, annual income, and occupation. Provide one new question about housing in Cook County that can be answered using at least one column of demographic data and at least one column of existing data and provide the type of analytical tool you would use to answer it.

Do assessment accuracy and potential over/under-valuation differ across owner race/ethnicity, gender, age, income, or occupation after holding property traits and location constant? I would model an assessment error/ratio = (Assessed or Prior Market Value—Sale Price)/Sale Price using a linear (or quantile) regression with demographic indicators as key predictors and controls for structure, time, and neighborhood fixed effects. Columns used: Sale Price, Estimate (Building), Estimate (Land) (or Prior Year Market Value Price), Building Square Feet, Land Square Feet, Age, Repair Condition, Central Air, Property Class, Neighborhood Code (or Census Tract), Sale Quarter, plus new demographic fields (race_ethnicity, gender, owner_age, annual_income, occupation).

0.5 Question 1e

Look at `codebook.txt` to see some of the unique regional features CCAO utilizes, such as **O'Hare Noise**. Now imagine you were in charge of predicting the **Sale Price** of houses in **your hometown** (your actual real life hometown/city - not the data provided). Propose a feature that you would want to collect specific to your location and hypothesize why it might be useful in predicting the sale price of houses.

Proposed hometown feature (Taiwan): MRT Accessibility Index.

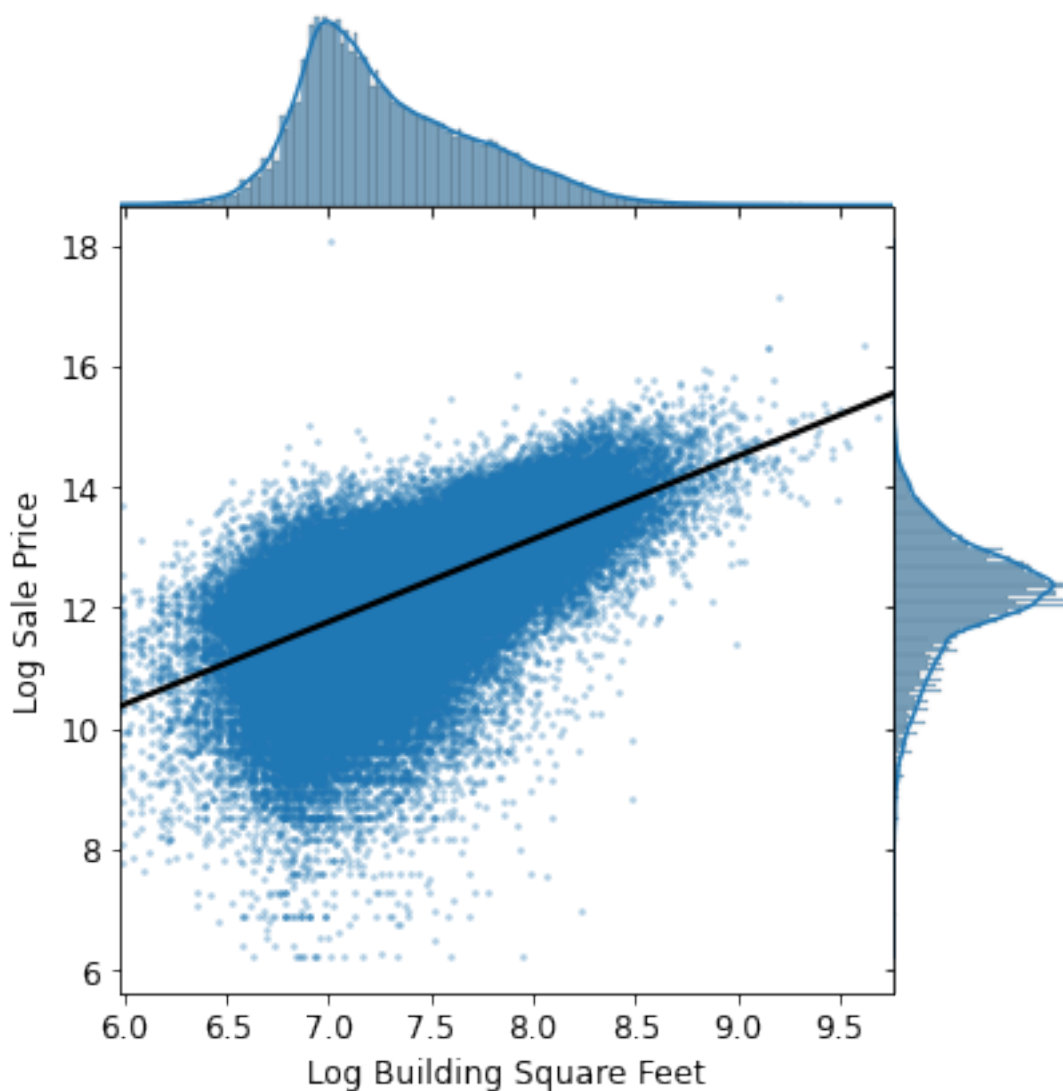
Collect a parcel-level score that combines (a) walking distance/time to the nearest MRT exit, (b) number of distinct lines reachable within 800 m, and (c) peak headway/frequency (from GTFS). Build it using NLSC/OSM pedestrian network routing for door-to-exit distance and Taipei/Kaohsiung Metro open data / MOTC PTX for station locations, exits, and schedules. Hypothesis: Higher MRT accessibility increases sale price (strong transit premium), with larger effects for smaller units and older buildings (parking scarce), and interactions with Road Proximity (transit substitutes car access) and Floor level/Elevator presence (walk-ups near MRT still command a premium due to commute savings).

0.6 Question 3b

In the visualization below, we created a `jointplot` with `Log Building Square Feet` on the x-axis, and `Log Sale Price` on the y-axis. In addition, we fit a simple linear regression line through the bivariate scatter plot in the middle.

Based on the following plot, would `Log Building Square Feet` make a good candidate as one of the features for our model? Why or why not?

Hint: To help answer this question, ask yourself: what kind of relationship does a “good” feature share with the target variable we aim to predict?



Yes—log Building Square Feet looks like a good predictor because it shows a clear, monotonic, roughly linear association with log Sale Price, which makes the relationship easy to model and interpret. The fitted line in the jointplot supports this. That said, if you read the scatter around that line as residuals, the slight downward drift at larger sizes hints at misspecification (diminishing returns/curvature or heteroskedasticity). I'd keep the feature in log form but consider adding mild nonlinearity (e.g., a spline or quadratic term), checking interactions with property class/age/condition, and using robust or clustered standard errors; these tweaks address the residual pattern without discarding a clearly informative feature.

0.7 Question 5c

Create a visualization that clearly and succinctly shows if there exists an association between **Bathrooms** and **Log Sale Price**. A good visualization should satisfy the following requirements: - It should avoid overplotting. - It should have clearly labeled axes and a succinct title. - It should convey the strength of the correlation between **Sale Price** and the number of rooms: in other words, you should be able to look at the plot and describe the general relationship between **Log Sale Price** and **Bathrooms**.

Hint 1: A direct scatter plot of the **Sale Price** against the number of rooms for all of the households in our training data might risk overplotting. Try it out and see why causes overplotting.

Hint 2: Take a closer look at the variable **Bathrooms**. What variable type is **Bathrooms** and what are the possible values **Bathrooms** can take? What visualizations might be good at showing this?

```
In [53]: df = (
    training_data.loc[training_data['Bathrooms'].between(0, 6), ['Bathrooms', 'Log Sale Price']]
    .dropna()
    .copy()
)
df['Bathrooms'] = df['Bathrooms'].astype(int)
order = sorted(df['Bathrooms'].unique())

# 2) Plot: boxplots by bathroom count + medians trend + sample sizes
fig, ax = plt.subplots(figsize=(8, 5))

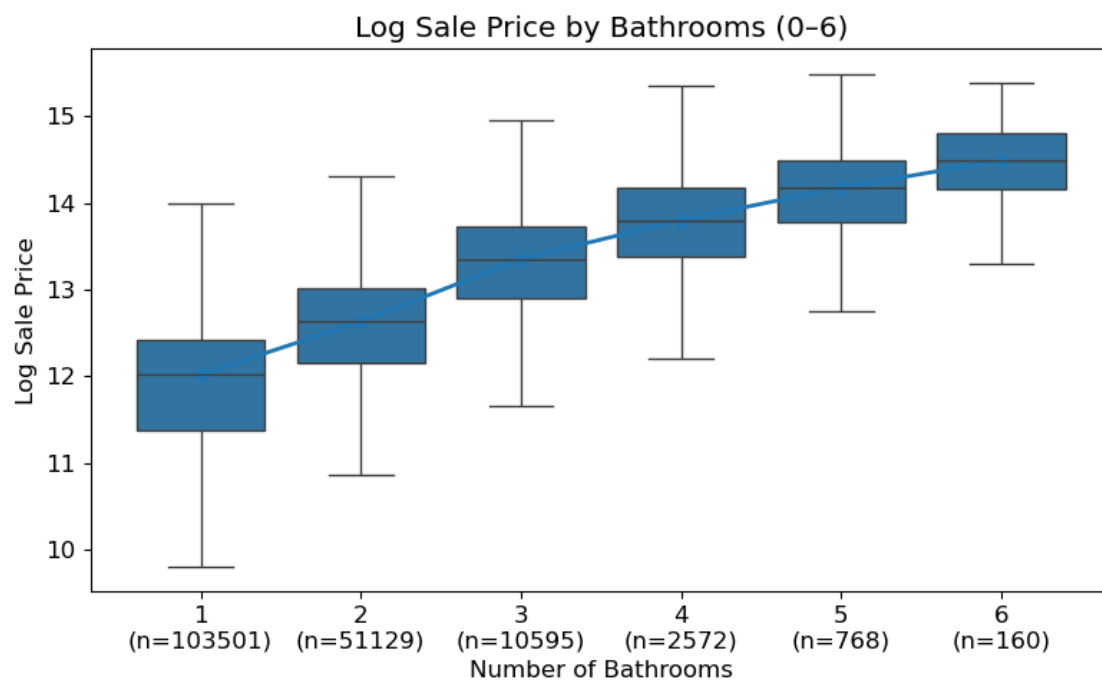
sns.boxplot(
    data=df,
    x='Bathrooms',
    y='Log Sale Price',
    order=order,
    whis=1.5,
    showfliers=False,
    ax=ax
)

# Overlay group medians to show the trend
medians = df.groupby('Bathrooms')['Log Sale Price'].median().reindex(order)
ax.plot(range(len(order)), medians.values, marker='o', linewidth=2)

# Annotate counts under each tick
counts = df['Bathrooms'].value_counts().reindex(order, fill_value=0)
ax.set_xticklabels([f"{b}\n(n={counts.loc[b]})" for b in order])

ax.set_xlabel("Number of Bathrooms")
ax.set_ylabel("Log Sale Price")
ax.set_title("Log Sale Price by Bathrooms (0-6)")
```

```
plt.tight_layout()
plt.show()
```



0.8 Question 8: Open-Ended Question

Welcome to another **open-ended question**.

If you have any feedback on this open-ended question, or any other homework question in Data 100, we encourage you to share your thoughts using the [content feedback form](#). You can also post to Ed.

Grading on open-ended questions is simple: **Clear evidence of thoughtfulness and effort will always receive full credit**. If your response is especially well-developed or creative, we may ask for permission to share it with the rest of the class so others can be inspired by your work! Underdeveloped ideas will receive half credit. Trivial or missing responses will receive no credit. We expect the vast majority of students to receive full credit.

SETUP: As mentioned in previous lectures, RMSE is a very powerful error metric. However, in [lecture 13](#) we learned that RMSE is not the best choice when evaluating the *fairness* of a property appraisal system and can be quite limiting. You can read about this in more detail [here](#) or rewatch [lecture 13](#). In short, RMSE fails to tell you:

1. The distribution of the errors
2. The sign (+/-) of the errors
3. The relative size of the errors

These aspects can be crucial when we are performing data analysis and developing a predictive model ourselves in this project.

TASK: Your task is to design your own custom error metric that may outperform RMSE in some of the aspects above to help you build a better machine learning model. Later, in Project A2 you will apply this custom error metric and compare it against RMSE

Your answer should consist of the following: 1. **Define a mathematical formula** for your custom error metric that penalizes unfair predictions 2. **Explain your reasoning** for why you think this metric captures fairness better than standard metrics like MSE or MAE 3. **Identify a potential counterargument** against your custom error metric. Why it might not be ideal in this context?

In the following cell (feel free to use $\LaTeX$ in combination with Markdown syntax), define a custom error metric that you believe would better capture fairness concerns in housing price prediction.

Important exception to existing course policies: **FOR THIS QUESTION ONLY** you are allowed to use large-language models (LLMs), like Gemini or ChatGPT. **However**, we strongly recommend thinking through this question on your own first and generating some different options. Only then you should use an LLM to discuss your possible metrics further. **If you copy-and-paste default output from an LLM on this question, there is a good chance that your submission will look identical or near-identical to many other students.** While we expect many answers to this question to have similarities, obvious default output will receive no credit. Spend time thinking about the presentation of your results.

Hint: If you are completely lost, you might consider metrics that weight errors differently based on property characteristics, or that penalize systematic biases across different groups.

Disclaimer: As Data Science students, you should be aware of important limitations and broader considerations when it comes to the use of LLMs. - LLMs do not guarantee factual accuracy and they are known to hallucinate (generate fabricated or misleading information). - LLMs are trained on large datasets that can reflect and reproduce biases in race, gender, culture, and ideology. - The use of LLMs may involve the sharing of sensitive and personal information.

IMPORTANT: If you have any questions, please read through the [FAQS](#) first. If you can't find the answer to your question there, feel free to ask your question on Ed.

I'm proposing a single fairness-aware loss that mixes robust relative error with group bias control. For each parcel I use the **signed relative error** $s_i = (\hat{y}_i - y_i)/y_i$ (so a 10% miss counts the same regardless of price). I pass $|s_i|$ through a **Huber** penalty $\phi(s_i) = s_i^2$ if $|s_i| \leq \tau$ and $\phi(s_i) = 2\tau|s_i| - \tau^2$ if $|s_i| > \tau$ to stay sensitive near zero but not let outliers dominate. Because over-assessment can raise a homeowner's tax bill, I weight errors asymmetrically with $\alpha(s_i) = \alpha_{\text{under}}$ for $s_i < 0$ and $\alpha(s_i) = \alpha_{\text{over}}$ with $\alpha_{\text{over}} > \alpha_{\text{under}}$ for $s_i \geq 0$. I also track systematic bias overall and within policy-relevant groups (e.g., neighborhoods): $\bar{s} = \frac{1}{n} \sum_i s_i$ and $b_k = \frac{1}{|g_k|} \sum_{i \in g_k} s_i$. The overall objective is $\mathcal{L} = (1 - \eta) \sqrt{\frac{1}{n} \sum_i \alpha(s_i) \phi(s_i)} + \eta (|\bar{s}| + \sum_k |b_k|)$ with small $\eta \in [0.1, 0.3]$, $\alpha_{\text{over}} > \alpha_{\text{under}}$, and τ around the 80–90th percentile of $|s_i|$. Compared to RMSE/MAE, this captures fairness better because it evaluates **relative** (not dollar) errors, keeps the **sign** so we can see and penalize over- vs. under-valuation, is **robust** to outliers via Huber, explicitly discourages **systematic group bias** through the $|b_k|$ terms, and encodes policy asymmetry against over-assessment. A fair counterargument is that the metric depends on design choices (which groups, η , λ , α , and τ), and small groups can make b_k noisy; to mitigate this I'd use predefined neighborhood codes, minimum group sizes or shrinkage (e.g., scale $|b_k|$ by $1/\sqrt{|g_k|}$), and report sensitivity analyses alongside results.